

TrES-3b

R-0479

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-3_180814 · created 2026-07-20T05:48:45+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2458344.8058 +/- 0.0014 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.102 +/- 0.042
Transit depth (Rp/Rs)^2	1.03 +/- 0.86 [%]
Orbital Inclination (inc)	86.59 +/- 2.65
Airmass coefficient 1 (a1)	1.019 +/- 0.017
Airmass coefficient 2 (a2)	-0.0087 +/- 0.0036
Scatter in the residuals of the lightcurve fit is	1.71 %
Best Comparison Star	#2 - [629, 328]
Optimal Aperture	3.5
Optimal Annulus	9.68
Transit Duration (day)	0.072 +/- 0.011

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.102 ± 0.042 vs 0.16309 (deviation 1.45σ)
Duration fit vs published	0.072 d vs 0.0567 d (deviation 1.39σ)
Mid-time O–C	-4.3 min
Depth SNR	2.4
Residual scatter	1.71 %

Light curve

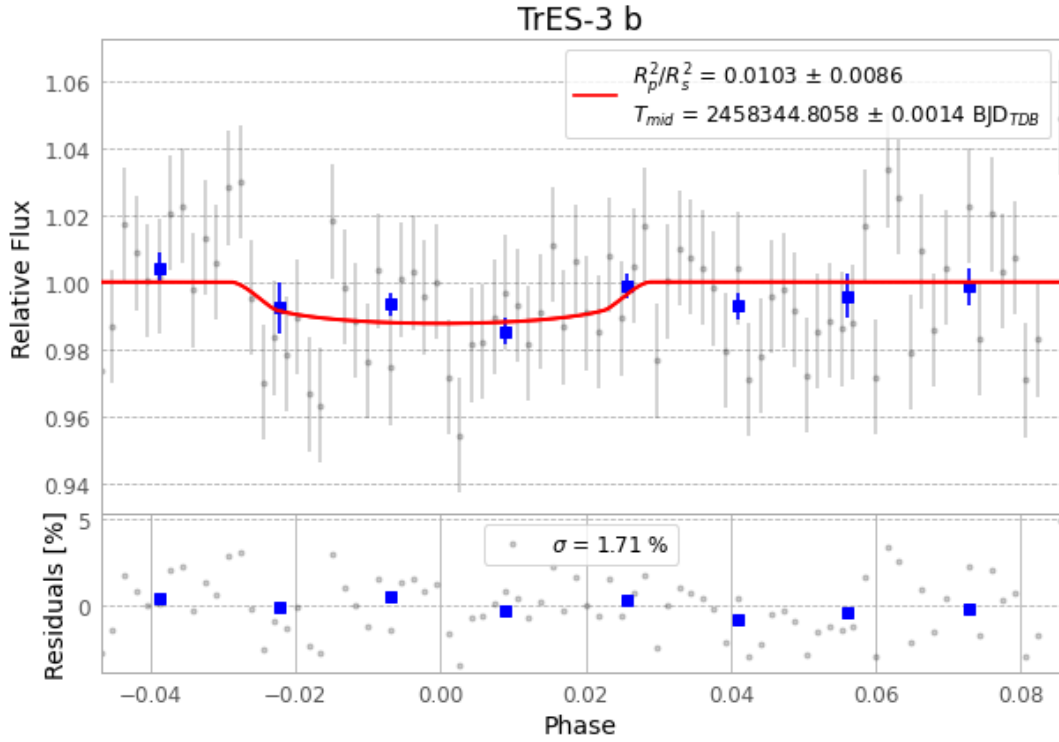


Figure 1 — FinalLightCurve_TrES-3 b_2018-08-13.png (SHA-256 d8cc0d20a149308c...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [328, 299], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 e96c89453aa92228...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 39d9b80

Data provenance

85 FITS frames (56 MB), TRES-3180814054812.FITS → TRES-3180814100012.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-3-180814.log (a167810406d2...), FinalLightCurve_TrES-3 b_2018-08-13.png (d8cc0d20a149...), FinalLightCurve_TrES-3 b_2018-08-13.csv (e1eb909d729b...)

Compute resources

Wall time 95.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-20 05:48Z.